

INNOVA JUNIOR COLLEGE

JC 2 PRELIMINARY EXAMINATIONS II

in preparation for General Certificate of Education Advanced Level

Higher 1

CANDIDATE
NAME

Civics Group

INDEX NUMBER

Mathematics

8864/01

Paper 1

19 Sep 2013

3 hours

Additional materials:

Answer Paper

Cover Page

List of Formulae (MF 15)

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your name, class and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **6** printed pages.



Section A: Pure Mathematics [35 marks]

- 1 Find, algebraically, the set of values of k for which

$$k^2 + kx + 2x - 1 - x^2 < 0$$

for all real values of x .

[4]

- 2 Given that $2e^x - 2e^{-x} = 3$, use the substitution $u = e^x$ to find the exact value of x .

[4]

Hence find the exact range of values of x such that $2e^x - 2e^{-x} > 3$.

[2]

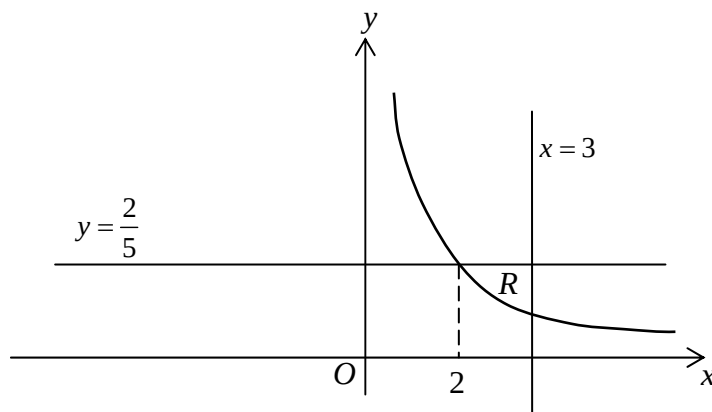
- 3 (i) Find $\frac{d}{dx} \left[\ln(3x^2 - 2) \right]$.

[2]

- (ii) Hence find $\int \frac{2x}{3x^2 - 2} dx$.

[2]

(iii)



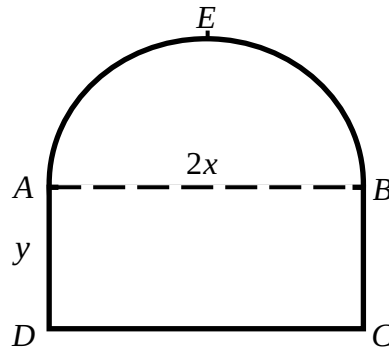
The diagram shows the region R in the first quadrant bounded by the curve

$y = \frac{2x}{3x^2 - 2}$, the lines $x = 3$ and $y = \frac{2}{5}$. Find the exact area of R , simplifying

your answer.

[3]

4



A window in a new building has the shape of a rectangle $ABCD$ joined to a semicircle AEB as shown in the diagram. It is given that $AB = 2x$ m and $AD = y$ m. The total perimeter $AEBCDA$ of the window is 8 m.

(i) Show that $y = \frac{8 - (2 + \pi)x}{2}$. [1]

(ii) Show that the area of the window is equal to $\left[8x - \left(\frac{4 + \pi}{2} \right) x^2 \right] \text{ m}^2$. [2]

(iii) Hence, by using differentiation, show that the maximum value of this area is $\frac{32}{4 + \pi} \text{ m}^2$. [5]

5 The curve C has equation $y = xe^{x^2} - 5x$.

(i) Sketch C , stating the coordinates of the points of intersection with the axes and the stationary points. [3]

(ii) Find the numerical value of the gradient of C at the point where $x = -1$. Give your answer correct to 4 decimal places. [1]

(iii) Hence find the equation of the tangent to C at the point where $x = -1$. Give your answer in the form $y = mx + c$, with m and c correct to 4 decimal places. [2]

(iv) This tangent meets the x -axis at P and the y -axis at Q . Find the coordinates of P and Q and hence find the area of triangle OPQ correct to 4 decimal places. [4]

Section B: Statistics [60 marks]

- 6** An educational institute offers a three-year diploma course. The Student Union of the institute would like to gather the students' opinions on the quality of the school's programmes and decides to use the lunch time period to survey a sample of 60 students at the canteen on a particular day.
- (i) Describe how the sample could be obtained using quota sampling and state, in context, a disadvantage that quota sampling has. [3]
 - (ii) Given that the ratio of students in Year 1, 2 and 3 is 5 : 4 : 3, describe how a more representative sample could be obtained. [2]
- 7** A sample of seven students from the same junior college is taken to find out how the ranking points (y) they obtained in their Mid-year Exam, conducted immediately after the June holidays, is correlated to the total amount of time (x hours) they spent on revision during the holidays. The results are given in the following table.

x	53	67	89	108	119	156	185
y	32	42	46	48	65	73	85

- (i) Give a sketch of the scatter diagram for the data, as shown on your calculator. [2]
- (ii) Find the product moment correlation coefficient and comment on its value in the context of this question. [2]
- (iii) Find the equation of the regression line of y on x in the form $y = a + bx$. Sketch this line on your scatter diagram. [2]
- (iv) Interpret, in context, the meaning of b . [1]
- (v) Estimate the ranking points of a student who spends 170 hours on revision, correct to the nearest integer value. Comment on the reliability of this estimate. [2]

- 8** Chest *A* contains 13 gold coins, 5 silver coins and 12 copper coins. Chest *B* contains 3 gold coins and 7 silver coins.

One of the chests is selected by throwing a fair die. If the die shows a number less than 3, chest *A* is selected and otherwise chest *B* is selected.

- (i) One coin is chosen at random from the selected chest and the colour of the coin is noted.
 - (a) Draw a tree diagram to represent this situation. [3]
 - (b) Find the probability that a silver coin is chosen. [2]
 - (c) Given that a silver coin is chosen, find the probability that it came from chest *A*. [2]
- (ii) Instead, two coins are chosen at random, without replacement, from the selected chest. Find the probability that both coins are of the same colour. [4]

- 9** In a national health survey, a sample of 20 elderly interviewees is obtained. The number of interviewees in the sample who are ‘at risk’ of having osteoporosis is denoted by X .

- (i) State, in context, what must be assumed for X to be well modelled by a binomial distribution. [2]

Assume now that X has the distribution $B(20, p)$.

- (ii) Given that $P(X=1)=0.00142$ correct to 5 decimal places, write down an equation for the value of p , and find this value numerically. Hence find $P(3 \leq X < 8)$. [4]

Suppose 60 samples of 20 interviewees each are taken.

- (iii) It is now given that $p = 0.290$. By using a suitable approximation, find the probability that at most 35 of such samples have more than 4 interviewees who are ‘at risk’ of having osteoporosis. [5]

- 10** In a country, the tourism board claims that the average spending of tourists visiting the country is \$ μ_0 per person. To test this claim, a random sample of 200 tourists is tracked and their spending per person, \$ x , are summarised by

$$\sum (x - 960) = -562 \text{ and } \sum (x - 960)^2 = 525250.$$

- (i) Find unbiased estimates of the population mean and variance. [3]
- (ii) A test carried out, at the 10% significance level, shows no significant evidence to reject the tourism board's claim. Using an algebraic method, find the range of possible values of μ_0 . [4]
- (iii) Explain, in context, the meaning of "at the 10% level of significance". [1]

After a series of promotional campaigns, the tourism board now claims that the average spending of tourists visiting the country is at least \$1100 per person. A new sample of 200 tourists is taken and the mean spending per person is found to be \$1093. Assuming that the population standard deviation is \$55, test the tourism board's new claim at the 3% significance level. [4]

- 11** A supermarket sells two types of guava, one from Thailand and one from Taiwan. The masses, in kilograms, of each type of guava have independent normal distributions. The means and standard deviations of these distributions, and the selling prices, in \$ per kilogram, are shown in the following table.

	Mean (kg)	Standard deviation (kg)	Selling price (\$ per kg)
Thailand	0.19	0.01	4.20
Taiwan	0.33	0.04	2.80

Stating clearly the mean and variance of all distributions that you use, find the probability that

- (i) the average mass of 8 randomly chosen Thai guavas is less than 0.18 kg, [2]
- (ii) the total mass of 6 randomly chosen Thai guavas is within 0.5 kg of the total mass of 5 randomly chosen Taiwanese guavas. [4]

Customer A buys 5 Thai guavas and 5 Taiwanese guavas. Customer B buys 10 Taiwanese guavas.

- (iii) Stating clearly the mean and variance of the distribution that you use, find the probability that customer A pays more than customer B. [6]